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Date: 08/28/26

IV. Part A – Analyze the System Before Coding

A. Actors

- Students
- Laboratory Staff

B. USE CASE

USE CASE 01

ITEM	DESCRIPTION
USE CASE	Request Borrow Equipment
PRIMARY ACTOR	Student
PRECONDITIONS	The student' requested equipment should exists and available, and if the student has not reached maximum numbers of borrowed equipment
MAIN ACTION	The student will request to borrow an equipment
EXPECTED RESULT	The request to borrow is approved and the system records the important details for borrowing the equipment
POSSIBLE FAILURE	Student is not allowed to borrow based on certain circumstances, equipment does not exist, equipment unavailable

USE CASE 02

ITEM	DESCRIPTION
USE CASE	Check Equipment Availability
PRIMARY ACTOR	Laboratory Staff
PRECONDITIONS	The student requested to borrow an

	equipment in the staff system
MAIN ACTION	The staff will check equipment availability
EXPECTED RESULT	The staff will allow the student to borrow the equipment if conditions are met
POSSIBLE FAILURE	A staff may decline a request if the equipment does not exist, or the student has reached maximum number of borrowed equipment

USE CASE 03

ITEM	DESCRIPTION
USE CASE	Return Equipment
PRIMARY ACTOR	Student
PRECONDITIONS	The student has an active borrowed equipment
MAIN ACTION	The student will return the equipment
EXPECTED RESULT	The equipment is returned and the equipment is marked as available again
POSSIBLE FAILURE	The student has no active borrowed equipment

C. Identify Domain Concepts

- **STUDENT**

- 1. What information must it contain?**

Important details of the students related to borrowing equipment such as name, id number, date borrowed, student borrowing status, equipment borrowed, and expected return date.

- 2. What rules or state belong to it?**

If the student is able to borrow based on his borrowing status.

- 3. What should ****not**** be the responsibility of that object?**

Any actions that a laboratory staff should do such as checking item availability, record student details, and managing the overall system a staff member is responsible for.

- **Equipment**

- 1. What information must it contain?**

Equipment type, availability, quantity, name, id, status

2. What rules or state belong to it?

Change its state between available and currently borrowed

3. What should ****not**** be the responsibility of that object?

Any responsibility of the student and staff.

- **Borrowing**

1. What information must it contain?

- EquipmentType, Availability, Quantity, Name, Id, Status, ExpectedReturnOn, BorrowedOn

2. What rules or state belong to it?

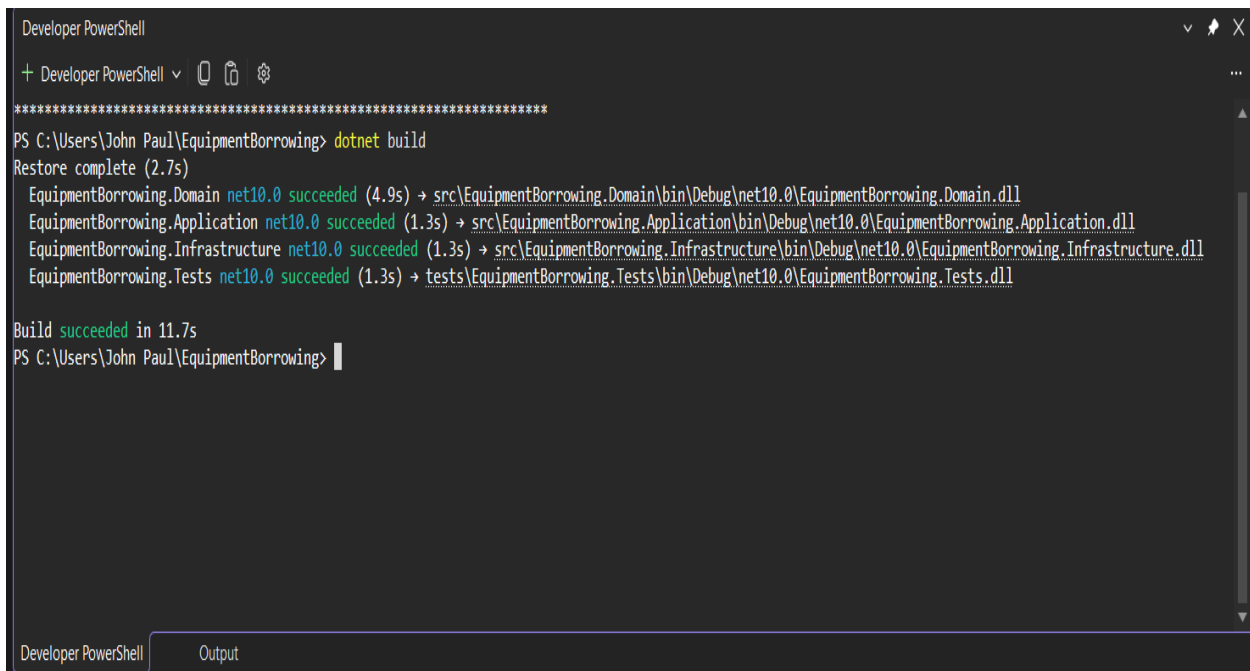
- It maintains the current state of the loan transaction (Active vs. Returned) and handles its own return logic.

3. What should ****not**** be the responsibility of that object?

- Any responsibilities from the actors like borrowing equipment, system handling, or fetching and saving data from storage etc.

SCREENSHOTS

Screenshot of the successful `dotnet build` output



```
Developer PowerShell
+ Developer PowerShell v [icon] [icon] [icon]
*****
PS C:\Users\John Paul\EquipmentBorrowing> dotnet build
Restore complete (2.7s)
EquipmentBorrowing.Domain net10.0 succeeded (4.9s) -> src\EquipmentBorrowing.Domain\bin\Debug\net10.0\EquipmentBorrowing.Domain.dll
EquipmentBorrowing.Application net10.0 succeeded (1.3s) -> src\EquipmentBorrowing.Application\bin\Debug\net10.0\EquipmentBorrowing.Application.dll
EquipmentBorrowing.Infrastructure net10.0 succeeded (1.3s) -> src\EquipmentBorrowing.Infrastructure\bin\Debug\net10.0\EquipmentBorrowing.Infrastructure.dll
EquipmentBorrowing.Tests net10.0 succeeded (1.3s) -> tests\EquipmentBorrowing.Tests\bin\Debug\net10.0\EquipmentBorrowing.Tests.dll

Build succeeded in 11.7s
PS C:\Users\John Paul\EquipmentBorrowing>
```

Output of one successful use case and one failure case

```
PS C:\Users\John Paul\Downloads\jamjam_fetch> dotnet test
Restore complete (1.7s)

Build succeeded in 2.8s
PS C:\Users\John Paul\Downloads\jamjam_fetch> dotnet run --project tests/EquipmentBorrowing.Tests
--- Success case ---
Borrowing #1 created: student 1 -> equipment 10, due 2026-09-04

--- Failure case: equipment already borrowed ---
Rejected as expected: Equipment 10 is not currently available.

--- Failure case: equipment does not exist ---
Rejected as expected: Equipment 999 does not exist.
PS C:\Users\John Paul\Downloads\jamjam_fetch>
```